

Syllabus for MA 563

Introduction to Differential Geometry

Course title from registrar

MA 563, section A1. Introduction to Differential Geometry.

Instructor

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Time and location

- Lecture times: TuTh 2:00pm–3:15pm
- Lecture location: CAS 214

Course summary

This is a proof-based introduction to differential geometry. We begin with the classical geometry of curves and surfaces in Euclidean space, developing the geometric meaning of curvature through concrete examples. We then study the intrinsic geometry of surfaces, including covariant differentiation, geodesics, parallel transport, and the Gauss–Bonnet theorem. In the final part of the course, we introduce smooth and Riemannian manifolds and explain how the geometry of surfaces extends to higher dimensions. Topics include, but are not limited to:

- regular curves, arclength, curvature, torsion, and the Frenet–Serret equations;
- parametrized surfaces, tangent planes, and the first fundamental form;
- the Gauss map, shape operator, principal curvatures, and the second fundamental form;
- Gaussian and mean curvature, ruled surfaces, and minimal surfaces;
- covariant differentiation, parallel transport, geodesics, and geodesic curvature;
- the Theorema Egregium, the Gauss–Codazzi equations, and the Gauss–Bonnet theorem;
- smooth manifolds, tangent spaces, Riemannian metrics, connections, and the curvature tensor.

Textbook

Wolfgang Kühnel, *Differential Geometry: Curves–Surfaces–Manifolds*, Third Edition, Student Mathematical Library, Volume 77, American Mathematical Society, 2015.

Prerequisites

Official course prerequisites: MA 225 (or MA 230), MA 226 (or MA 231), and MA 242 (or MA 442). Students should be comfortable with multivariable calculus and linear algebra. The course relies on rigorous mathematical proof, so prior experience reading and writing proofs is strongly recommended.

Problem sets

An ungraded problem set will be distributed approximately every two weeks. Each set will ordinarily contain about five to seven problems. These problems are an essential source of practice: the examinations will be at a comparable level. Students are strongly encouraged to discuss the problems with one another, but should make sure that they can write and explain complete solutions independently. Problem-set solutions and/or discussion of selected problems will be provided.

If you are struggling to solve the homework problems then please make sure to ask questions and/or attend office hours. The examinations will cover the same material as the homework sets and will be at a comparable level.

Assessment

- **Midterm exams.** There will be two in-class midterm exams.

- Midterm exam 1: **Thursday, October 8.**
- Midterm exam 2: **Thursday, November 19.**

Each midterm exam must be completed individually and without any aid from devices, notes, or textbooks. Students can use one “formula sheet”. There are **no make-up midterm exams** under any circumstances outside of medical emergencies.

- **Group project.** Students will work in groups of two on an extension of material from the course. Each group will submit a mathematical expository paper and give a short presentation. The project expectations, suggested topics, and rubric appear in the separate project handout. Project deadlines and presentation dates are TBA.
- **Final exam.** There will be a cumulative two-hour final exam occurring during the university-wide final-examination period whose date and time follow the *Final Exam Matrix* posted by the registrar. According to this matrix, the date and time for our final exam is:

Tuesday, December 15 | 3:00pm–5:00pm

Grade breakdown

First midterm	20%
Second midterm	20%
Group project	25%
Final exam	35%
Total	100%

The biweekly problem sets are ungraded and do not contribute directly to the numerical course grade.

How to succeed in this class

- Attend class and work steadily. Differential geometry combines geometric intuition, calculation, and proof; each of these improves through regular practice.
- Treat the ungraded problem sets as required preparation. Attempt problems before consulting solutions, and practice writing complete arguments without notes.
- If you are struggling, reach out by email or attend office hours early in the semester. October 7 is the last day to drop this class without a "W" grade. November 6 is the last day to drop this class with a "W" grade.
- Additional academic-support resources are available through the Department of Mathematics and Statistics and the Educational Resource Center; current details are available at <https://www.bu.edu/math/> and <https://www.bu.edu/erc/>.

This syllabus is subject to minor changes. Any change made after the first day of class will be announced and reflected in the updated syllabus.

Tentative 25-meeting lecture plan

The final examination is outside these 25 class meetings. The two midterms are included as Meetings 9 and 20. Section references are approximate and may shift according to the pace of the class.

Part I. Curves in Euclidean space

Meeting 1. Review of calculus and Euclidean geometry.

Kühnel: Chapter 1. A review of the analytical tools used throughout the course: Euclidean inner products and norms, differentiable maps, derivatives as linear transformations, Jacobian matrices, and the chain rule. For a smooth map

$$F : \mathbf{R}^n \longrightarrow \mathbf{R}^m,$$

the derivative at p is the linear map

$$dF_p : \mathbf{R}^n \longrightarrow \mathbf{R}^m.$$

We recall the inverse and implicit function theorems, regular level sets, and the existence and uniqueness principle for ordinary differential equations. In particular,

$$F^{-1}(c)$$

is locally a smooth submanifold near p when dF_p has the appropriate rank.

Meeting 2. Regular curves, arclength, and plane curvature.

Kühnel: §§2A–2B. Regular curves

$$c : I \longrightarrow \mathbf{R}^n, \quad c'(t) \neq 0,$$

reparametrization, speed, length, and arclength:

$$L(c) = \int_a^b \|c'(t)\| dt, \quad s(t) = \int_{t_0}^t \|c'(u)\| du.$$

For a unit-speed plane curve, define

$$T = c', \quad T' = \kappa N.$$

Signed curvature, orientation, and osculating circles.

Meeting 3. Space curves, curvature, and torsion.

Kühnel: §§2B–2C. For a unit-speed curve in $\mathbf{R}^3$, define

$$T = c', \quad \kappa = \|T'\|, \quad N = \frac{T'}{\kappa}, \quad B = T \times N.$$

The torsion

$$\tau = -\langle B', N \rangle$$

measures the failure of the curve to remain in its osculating plane. We compute curvature and torsion for standard examples.

Meeting 4. Frenet equations and the fundamental theorem of curves.

Kühnel: §§2C–2D. Derive the Frenet–Serret equations

$$\begin{aligned} T' &= \kappa N, \\ N' &= -\kappa T + \tau B, \\ B' &= -\tau N. \end{aligned}$$

Curvature and torsion determine a unit-speed space curve up to an orientation-preserving Euclidean motion.

Part II. The local theory of surfaces

Meeting 5. Parametrized surfaces and tangent planes.

Kühnel: §3A. Surface patches

$$x : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3, \quad x_u \times x_v \neq 0,$$

coordinate changes, tangent vectors, tangent planes, and examples including graphs and surfaces of revolution.

Meeting 6. The first fundamental form.

Kühnel: §3A. The induced metric is

$$I = E du^2 + 2F du dv + G dv^2,$$

where

$$E = \langle x_u, x_u \rangle, \quad F = \langle x_u, x_v \rangle, \quad G = \langle x_v, x_v \rangle.$$

Lengths, angles, and surface area:

$$dA = \sqrt{EG - F^2} du dv.$$

Meeting 7. The Gauss map and the second fundamental form.

Kühnel: §3B. For an oriented surface, define

$$N = \frac{x_u \times x_v}{\|x_u \times x_v\|}, \quad S = -dN.$$

The shape operator S and second fundamental form

$$II(v, w) = \langle Sv, w \rangle$$

measure the extrinsic bending of the surface.

Meeting 8. Principal, Gaussian, and mean curvature.

Kühnel: §3B. Principal curvatures are the eigenvalues k_1, k_2 of S . Define

$$K = k_1 k_2 = \det S, \quad H = \frac{k_1 + k_2}{2} = \frac{1}{2} \operatorname{tr} S.$$

Principal curvatures are the eigenvalues k_1, k_2 of S . Define

$$K = k_1 k_2 = \det S, \quad H = \frac{k_1 + k_2}{2} = \frac{1}{2} \operatorname{tr} S.$$

Euler's formula for normal curvature and the classification of elliptic, hyperbolic, parabolic, and planar points.

Meeting 9. Midterm examination I.

The first in-class midterm covers Meetings 1–8.

Part III. Special surfaces and intrinsic geometry

Meeting 10. Surfaces of revolution and ruled surfaces.

Kühnel: §3C. Surfaces of revolution

$$x(u, v) = (r(u) \cos v, r(u) \sin v, z(u))$$

and ruled surfaces

$$x(u, v) = c(u) + v d(u).$$

Examples include cylinders, cones, helicoids, and developable surfaces.

Meeting 11. Minimal surfaces.

Kühnel: §3D. A surface is minimal when

$$H = 0.$$

Examples include the plane, catenoid, and helicoid. We relate mean curvature to the first variation of area:

$$\left. \frac{d}{dt} \right|_{t=0} A(S_t) = -2 \int_S H \varphi dA.$$

Meeting 12. Covariant differentiation on a surface.

Kühnel: §4A. For tangent vector fields X, Y , define the covariant derivative by tangential projection:

$$\nabla_X Y = (D_X Y)^\top.$$

Metric compatibility, the product rule, and differentiation of vector fields along curves.

Meeting 13. Parallel transport and geodesics.

Kühnel: §4B. A vector field V along c is parallel when

$$\nabla_{\dot{c}} V = 0.$$

A curve is geodesic when

$$\nabla_{\dot{c}} \dot{c} = 0.$$

Examples on planes, cylinders, spheres, and surfaces of revolution.

Meeting 14. Geodesic equations and geodesic curvature.

Kühnel: §4B. In local coordinates, the geodesic equations are

$$\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0.$$

For a unit-speed surface curve,

$$c'' = k_n N + k_g n_g.$$

Thus $k_g = 0$ precisely when the curve is geodesic.

Meeting 15. The Gauss equation and Theorema Egregium.

Kühnel: §4C. The Gauss equation relates intrinsic curvature to the shape operator. In coordinates,

$$K = \frac{LN - M^2}{EG - F^2}.$$

Gauss's Theorema Egregium states that K is determined entirely by the first fundamental form and is therefore preserved by local isometries.

Meeting 16. Gauss–Codazzi and the fundamental theorem of surfaces.

Kühnel: §§4C–4D. The compatibility equations include

$$(\nabla_X S)(Y) = (\nabla_Y S)(X).$$

Compatible first and second fundamental forms determine a surface locally, uniquely up to a rigid motion.

Meeting 17. Gaussian curvature in special coordinates.

Kühnel: §4E. Geodesic, orthogonal, and conformal coordinates. For a conformal metric

$$g = e^{2\lambda}(du^2 + dv^2),$$

the Gaussian curvature is

$$K = -e^{-2\lambda}(\lambda_{uu} + \lambda_{vv}).$$

Meeting 18. The local Gauss–Bonnet theorem.

Kühnel: §4F. For a suitable region R with piecewise smooth boundary,

$$\int_R K \, dA + \int_{\partial R} k_g \, ds + \sum_i \theta_i = 2\pi \chi(R).$$

Orientation, exterior angles, and geodesic triangles.

Meeting 19. Global Gauss–Bonnet and applications.

Kühnel: §§4F–4G. For a compact oriented surface without boundary,

$$\boxed{\int_S K \, dA = 2\pi \chi(S).}$$

For a closed orientable surface of genus g ,

$$\int_S K \, dA = 4\pi(1 - g).$$

Applications connecting curvature, topology, triangulations, and vector fields.

Meeting 20. Midterm examination II.

The second in-class midterm covers Meetings 10–19, with any cumulative material announced in advance.

Part IV. An introduction to Riemannian manifolds

Meeting 21. Smooth manifolds and coordinate charts.

Kühnel: §5A. Smooth n -manifolds, coordinate charts

$$\varphi : U \subset M \longrightarrow \mathbf{R}^n,$$

atlases, transition maps, and smooth maps between manifolds. Examples include spheres, projective spaces, and matrix groups.

Meeting 22. Tangent spaces and differentials.

Kühnel: §5B. Equivalent descriptions of $T_p M$ using velocities of curves, coordinate vectors, and derivations. For a smooth map $F : M \rightarrow N$,

$$dF_p : T_p M \longrightarrow T_{F(p)} N.$$

Meeting 23. Riemannian metrics.

Kühnel: §5C. A Riemannian metric is a smoothly varying inner product

$$g_p : T_p M \times T_p M \longrightarrow \mathbf{R}.$$

It determines lengths and energies:

$$L(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma})} dt, \quad E(\gamma) = \frac{1}{2} \int_a^b g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}) dt.$$

Meeting 24. The Levi-Civita connection on a manifold.

Kühnel: §5D. Existence and uniqueness of the torsion-free, metric-compatible connection:

$$\nabla_X Y - \nabla_Y X = [X, Y],$$

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

Geodesics and the exponential map

$$\exp_p(v) = \gamma_v(1).$$

Meeting 25. Curvature beyond surfaces.

Kühnel: selected material from §§6A–6B. Introduce the Riemann curvature tensor

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

and sectional curvature

$$K(\sigma) = \frac{\langle R(u, v)v, u \rangle}{\langle u, u \rangle \langle v, v \rangle - \langle u, v \rangle^2}.$$

This provides a synthesis of surface curvature and its higher-dimensional generalization. Project presentation scheduling: TBA.

OLD - based on do Carmo

Tentative 25-meeting lecture plan

The final examination is outside these 25 class meetings. The two midterms are included as Meetings 9 and 20. This schedule is tentative and may shift in response to the pace of the class.

Part I. Curves in $\mathbb{R}^3$

Meeting 1. Parametrized curves and arclength.

Do Carmo: §§1-2, 1-3. Regular parametrized curves $\alpha : I \rightarrow \mathbb{R}^3$, reparametrization, speed, length, and local arclength parameters:

$$L(\alpha) = \int_a^b \|\alpha'(t)\| dt, \quad s(t) = \int_{t_0}^t \|\alpha'(u)\| du.$$

Meeting 2. Curvature and the Frenet frame.

Do Carmo: §§1-4, 1-5. For a unit-speed curve, define

$$T = \alpha'(s), \quad \kappa = \|T'(s)\|, \quad N = \frac{T'}{\kappa}, \quad B = T \times N.$$

We interpret curvature using osculating circles and compute examples.

Meeting 3. Torsion and the Frenet–Serret equations.

Do Carmo: §1-5. Define torsion and derive

$$\begin{aligned} T' &= \kappa N, \\ N' &= -\kappa T + \tau B, \\ B' &= -\tau N. \end{aligned}$$

We discuss the geometric meaning of torsion and the characterization of planar curves by $\tau = 0$.

Meeting 4. The fundamental theorem of curves.

Do Carmo: §§1-5, 1-6; *selected material from §1-7*. Curvature and torsion determine a unit-speed space curve up to an orientation-preserving rigid motion:

$$(\kappa(s), \tau(s)) \iff \alpha(s) \text{ up to rigid motion.}$$

Local canonical form and selected global facts about plane curves.

Part II. Regular surfaces and the metric

Meeting 5. Regular surfaces and examples.

Do Carmo: §2-2. Parametrized surfaces $x : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, the regularity condition

$$x_u \times x_v \neq 0,$$

graphs, level surfaces, surfaces of revolution, and coordinate neighborhoods.

Meeting 6. Change of parameters, tangent planes, and differentials.

Do Carmo: §§2-3, 2-4. Smooth functions and maps on surfaces, coordinate changes, tangent vectors, tangent planes, and the differential

$$dF_p : T_p S \longrightarrow T_{F(p)} \widetilde{S}.$$

Meeting 7. The first fundamental form.

Do Carmo: §2-5. The induced metric

$$I = E du^2 + 2F du dv + G dv^2,$$

where $E = \langle x_u, x_u \rangle$, $F = \langle x_u, x_v \rangle$, and $G = \langle x_v, x_v \rangle$. Lengths, angles, and areas from I .

Meeting 8. Area and orientation.

Do Carmo: §§2-5, 2-6; *selected material from* §2-7. Surface area,

$$A(x(U)) = \iint_U \|x_u \times x_v\| du dv = \iint_U \sqrt{EG - F^2} du dv,$$

orientability, unit normal fields, and examples including the Möbius strip.

Meeting 9. Midterm examination I.

The first in-class midterm covers Meetings 1–8. Date: TBA.

Part III. Extrinsic curvature and the Gauss map

Meeting 10. The Gauss map and the shape operator.

Do Carmo: §3-2. For an oriented surface with Gauss map $N : S \rightarrow S^2$, define the shape operator

$$S_p = -dN_p : T_p S \longrightarrow T_p S$$

and the second fundamental form $II_p(v, w) = \langle S_p v, w \rangle$.

Meeting 11. Principal curvatures and Euler's formula.

Do Carmo: §3-2. Self-adjointness of the shape operator, principal curvatures k_1, k_2 , principal directions, and normal curvature

$$k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta.$$

Meeting 12. Gaussian and mean curvature.

Do Carmo: §§3-2, 3-3. Define

$$K = \det S = k_1 k_2, \quad H = \frac{1}{2} \operatorname{tr} S = \frac{k_1 + k_2}{2},$$

and classify elliptic, hyperbolic, parabolic, and planar points.

Meeting 13. Further geometry of the Gauss map.

Do Carmo: §3-3. Asymptotic and conjugate directions, lines of curvature, curvature of graphs, and selected global consequences of the Gauss map.

Meeting 14. Ruled, developable, and minimal surfaces.

Do Carmo: §3-5. Ruled surfaces and the condition $K = 0$ for developables; minimal surfaces and the condition

$$H = 0.$$

Examples may include the cylinder, cone, helicoid, and catenoid.

*Part IV. Intrinsic geometry***Meeting 15. Vector fields and covariant differentiation.**

Do Carmo: §3-4 and related material in §4-4. Tangent vector fields, derivatives along curves, and the Levi-Civita covariant derivative as tangential projection:

$$\nabla_X Y = (D_X Y)^\top.$$

Meeting 16. Local isometries and conformal maps.

Do Carmo: §4-2. Intrinsic quantities and maps preserving the first fundamental form:

$$\langle dF_p(v), dF_p(w) \rangle = \langle v, w \rangle.$$

Examples include the local isometry between a cylinder and a plane.

Meeting 17. Christoffel symbols and the Theorema Egregium.

Do Carmo: §4-3. The coordinate formula

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$$

and Gauss's theorem that K is determined entirely by the first fundamental form. Consequently, local isometries preserve Gaussian curvature.

Meeting 18. Gauss–Codazzi and the fundamental theorem of surfaces.

Do Carmo: §4-3. The compatibility equations for I and II , including

$$(\nabla_X S)(Y) = (\nabla_Y S)(X),$$

and the local existence and uniqueness of a surface with prescribed compatible fundamental forms.

Meeting 19. Geodesics and the geodesic equations.

Do Carmo: §4-4. A curve γ is geodesic when

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0, \quad \text{equivalently} \quad \ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0.$$

Local existence and uniqueness, with examples on planes, cylinders, spheres, and surfaces of revolution.

Meeting 20. Midterm examination II.

The second in-class midterm covers Meetings 10–19, with any cumulative material announced in advance. Date: TBA.

Meeting 21. Parallel transport and geodesic curvature.

Do Carmo: §4-4. Parallel vector fields along curves, holonomy, and the decomposition for a unit-speed surface curve

$$\gamma'' = k_n N + k_g n_g.$$

In particular, $k_g = 0$ characterizes geodesics.

Meeting 22. Geodesics as critical and locally minimizing curves.

Do Carmo: §4-4 and the relevant later material in Chapter 4. Energy and length, the first variation, and the distinction between local and global length minimization. Short radial geodesics minimize length in a normal neighborhood.

Part V. Global curvature

Meeting 23. The local Gauss–Bonnet theorem.

Do Carmo: §4-5. For a suitable region R with piecewise smooth boundary,

$$\int_R K \, dA + \int_{\partial R} k_g \, ds + \sum_i \theta_i = 2\pi \chi(R),$$

with careful attention to orientation and exterior angles.

Meeting 24. Global Gauss–Bonnet and applications.

Do Carmo: §4-5 and selected material from Chapter 5. For a compact oriented surface without boundary,

$$\boxed{\int_S K \, dA = 2\pi \chi(S).}$$

For a closed orientable surface of genus g , this becomes

$$\int_S K \, dA = 4\pi(1 - g).$$

Applications connect curvature, topology, triangulations, and the geometry of compact surfaces.

Meeting 25. The exponential map and constant-curvature geometry.

Do Carmo: relevant later sections of Chapter 4 and selected material from Chapter 5. The exponential map

$$\exp_p : T_p S \supset U \longrightarrow S, \quad \exp_p(v) = \gamma_v(1),$$

normal neighborhoods, and the model geometries $K = 0$, $K > 0$, and $K < 0$. This meeting also provides a synthesis of the course. Project presentation scheduling: TBA.